

Quantitative emergence and finite-time startup of productive polymer reactors: stochastic operating guarantees and certified static availability bounds

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Abstract

A reflexively autocatalytic and food-generated (RAF) set certifies that a reaction network could regenerate its catalysts from food. It supplies no deadline, stock requirement, export quota, or reliability for a finite reactor started from food alone. We study the gap between these two kinds of statement in the reversible binary polymer model with a capped Zipf catalysis source, in a fed and diluted stochastic mass-action reactor with count scale V . The operating requirement is a single uninterrupted trajectory that holds stock of a length-four species at times 1, 100 and 199, exports more than $V/10$ nonfood monomer equivalents in each of the windows $(1, 100]$ and $(100, 199]$, stays in a mass corridor, and consumes at most a stated food supply. For maximum word length $n \geq 4$ and integer $V \geq \max\{10^{24}, 10^{13}n\}$ we prove that every environment in an explicit self-catalytic witness class, with every other nonfood catalytic row unrestricted and every admitted kinetic coefficient, completes this requirement with probability at least 0.9989. The earlier sufficient scale for the same event, model and source was $10^{60}(n+1)^2$, or about 5×10^{49} after retuning. The gain comes from changing the observable: a guarded count-tail estimate supplies stock, and a one-sided logarithmic reward that discards favourable product fluctuations forces nonfood occupation and hence export. The same good event yields export above $0.42V$ over $(1, 199]$, recovery of at least $1/5700$ of supplied monomer, an explicit deadline–quota–reliability trade-off, and a conditional reliability above $1 - 1.7 \times 10^{-7}$ for the original requirement. Averaging over the source gives $\Omega(2^{-n})$ success probability along the critical Zipf sequence. For the separate independent static channel fields that govern RAF emergence in the critical window, embedded dimer and three-type prefix branching processes give two-sided survival certificates, including enclosures of width below 0.006 at openness $1/4$ and below 10^{-9} at openness $1/2$ in both the split and the quotient channel conventions. The finite stochastic theorem, the dimer certificates and the inverse certificate are machine-checked in Lean 4; every other statement is proved conventionally in the text and labelled as such.

1 Introduction

Collectively autocatalytic sets were proposed by Kauffman [12, 21] as a route by which a random chemistry could sustain itself without a single self-replicating molecule. Their combinatorial formalization, the reflexively autocatalytic and food-generated set or RAF [16, 27, 28], is a closure statement: a set of reactions can produce all of its reactants from a food set, and each of its reactions has a catalyst so produced. In the binary polymer model RAFs appear once the mean number of reactions catalysed per molecule grows linearly with the maximum word length [20, 22], and heterogeneous, power-law catalysis changes both their appearance and their size [18, 19].

A closure statement is not an operating guarantee. It supplies no initial state, no kinetic law, no deadline, no collection rule, and no resource account. Kinetic studies of autocatalytic sets are as old as the model [12]. Hordijk and Steel simulated molecular flow in RAF networks [17]; Giri and Jain separated the catalytic strength needed to maintain a large-molecule state from that needed to reach it from food [14]; Ravoni studied how composition shapes stochastic dynamics [26]; the stoichiometric theory of autocatalysis classifies motifs that can amplify their own species [7, 30]; and seed-dependent activation shows why food-only initiation is a substantive restriction [24]. A companion paper [4] fixed one random model in which the question can be posed exactly: the capped-Zipf binary polymer model in a fed, diluted, stochastic mass-action reactor started from food, with a two-window operating requirement on one uninterrupted trajectory. It proved a positive lower bound on success, a matching converse, and the rarity order $\Theta(2^{-n})$ along the critical Zipf sequence. Its sufficient count scale, however, was $10^{60}(n+1)^2$, or about 5×10^{49} after separate retuning of its tolerances, against a necessary first-birth scale of 1.43×10^7 .

This paper makes the operating guarantee quantitative. The model, the source, the kinetic coefficients, the preparation and the operating requirement are unchanged; only the proof changes. We ask three questions. How large must a reactor be before a specified autocatalytic witness completes the requirement with high, explicit probability? What does the same argument say about output, deadlines and material recovery? And, for the separate static probability spaces that govern the *availability* of autocatalytic organizations in the critical window [1–3], can survival probabilities at moderate openness be certified cheaply to useful precision?

Contributions.

- (i) *A finite startup theorem at a much smaller scale* (Theorem 3.1). For every $n \geq 4$, every Zipf exponent $\alpha > 1$ and every integer $V \geq V_*(n) = \max\{10^{24}, 10^{13}n\}$, every environment of the witness class, however its other nonfood catalytic rows are filled, and every admitted kinetic coefficient family, completes the original two-window requirement with probability at least 9989/10000. Averaging over the source gives the exact witness mass times 9989/10000, and every successful trajectory has net internal nonfood synthesis above $V/5$. [Lean]
- (ii) *A mechanism*. The earlier scale was set by an absolute tolerance, about 10^{-21} , on the selected product coordinate, squared inside a concentration estimate; helpful product births inflated a symmetric fluctuation budget. We replace it by a guarded count-tail estimate with two boundary states (Section 4.3) and by a retained logarithmic reward that keeps every adverse product fluctuation and discards favourable ones (Section 4.4). Section 4.8 identifies which inequality sets each of 10^{24} and $10^{13}n$.
- (iii) *All windows on one event* (Theorem 3.2). On a single event of outer failure probability below 11/10000, every window $(s, t] \subseteq (1, 199]$ exports more than $q(t-s)V$ with $q(d) = (\frac{19}{10}d - 82)/624 - \frac{1}{20}$. [Lean]
- (iv) *Consequences* (Section 5). Export above $263V/624 > 0.42V$ over $(1, 199]$ and recovery above $1/5700$ of all supplied monomer; any window longer than $1756/19 \approx 92.4$ residence times exports more than $V/10$; the true failure budget at V_* is below 9.1×10^{-5} , and with a different noise level the original requirement has conditional reliability above $1 - 1.7 \times 10^{-7}$ (Proposition 5.1); 1600 independent redraws of chemistry suffice for a 95% screening guarantee at $n = 4$; $\liminf_n 2^n P_n \geq 0.9989 (6/\pi^2)^6 9/(2\pi^2)$ along the critical sequence; and a finite-horizon continuity bound under weak food catalysis. [conventional]
- (v) *Static availability certificates* (Theorem 3.4). For the independent static channel fields of the critical-window law, an injectively embedded dimer forest gives $S_\nu(a) \geq 1 - r^{16}$ whenever

	Dynamic model (Sections 2.1–4)	Static model (Sections 2.4, 6)
Random object	catalytic rows C_z with capped Zipf degrees; kinetic marks	independent Bernoulli(a) openness of every channel
Dependence	strong within a row; rows independent	none between channel coordinates
Words	length at most n	all finite lengths
State	molecule counts, continuous time	set of available word types
Event	two-window operating requirement by time 199	unbounded ordinary reversible closure
Parameter	Zipf exponent $\alpha > 1$, count scale V	openness $a \in [0, 1]$
Probability	$\mathbb{P}_\omega(F_{n,V})$ and its source average	$S_{\text{sp}}(a), S_{\text{qt}}(a)$

Table 1: The two probability spaces. No statement below multiplies probabilities from different columns.

$1 - a + ar^4 \leq r$, [Lean] and an embedded three-type prefix branching process gives sharper bounds that reach above 99% at $a = 1/4$, where the dimer construction is critical. [conventional] Closed food gateways give $S_\nu(a) \leq 1 - (1 - a)^{g_\nu}$. The resulting enclosures have width below 0.006 at $a = 1/4$, below 3×10^{-4} at $a = 3/10$ and below 10^{-9} at $a = 1/2$ (Table 2).

Three kinds of evidence. Statements marked [Lean] have a named declaration in a Lean 4 [10] and Mathlib [29] development with a strict compiler receipt (Appendix A). Statements marked [conventional] are proved in full in the text and have no separate formal export; several are short deductions from formal statements. *Exact arithmetic* (rational enclosures and integer powers) is replayed by a script distributed with the source; *numerical diagnostics* are labelled as such and never used as premises.

What is not claimed. The chemistry is schematic: binary words, ordered split identities, one basal scale, paired reversible coefficients, uniform washout, and a well-mixed vessel. The count scale is sufficient, not a threshold, and remains conservative by many orders of magnitude (Section 7). The witness is a single self-catalytic ligation embedded in an arbitrary background, not a collectively autocatalytic cycle. Output is aggregate nonfood monomer effluent, not a purified or functional product. No matching upper bound, selection theorem or reliability tending to one is transferred to the new scale (Section 5.4). The static probabilities concern a different random experiment from the dynamic source and are never multiplied with dynamic probabilities.

Organization. Section 2 fixes both models. Section 3 states the main results. Section 4 proves the dynamic theorems, Section 5 their consequences, and Section 6 the static certificates. Section 7 discusses scale and interpretation, Section 8 the diagnostics that shaped the proof, and Section 9 open problems. Appendix A maps statements to evidence.

2 Two models and their probability spaces

2.1 Catalogue, source and kinetics

Fix $n \geq 4$. Let $\mathcal{X}_n$ be the nonempty binary words of length at most n , and let $\mathcal{R}_n$ be the ordered split identities $r = (u, v, uv)$ with $|u|, |v| \geq 1$ and $|uv| \leq n$. Then

$$|\mathcal{X}_n| = 2^{n+1} - 2, \quad R_n = |\mathcal{R}_n| = \sum_{\ell=2}^n (\ell-1)2^\ell = (n-2)2^{n+1} + 4. \quad (2.1)$$

Each identity has a ligation direction $u + v \rightarrow uv$ and a cleavage direction $uv \rightarrow u + v$. The food set is $\mathcal{F} = \{0, 1, 00, 01, 10, 11\}$.

Source. For a real exponent $\alpha > 1$, independently for each molecule $z \in \mathcal{X}_n$ draw $K_z \geq 1$ with $\mathbb{P}(K_z = k) = k^{-\alpha}/\zeta(\alpha)$, set $D_z = \min(K_z, R_n) - 1$, and given $D_z = d$ let the row $C_z \subseteq \mathcal{R}_n$ of identities catalysed by z be a uniformly random d -subset. The whole Zipf tail sits at the capped degree $R_n - 1$; truncating and renormalizing would define a different source. Rows of different molecules are independent; assignments within a row are strongly dependent. Put

$$e_\alpha = \mathbb{P}(D = 0) = \frac{1}{\zeta(\alpha)}, \quad p_{\alpha,n} = \mathbb{P}(r \in C_z) = \frac{\mathbb{E}D}{R_n}. \quad (2.2)$$

Kinetic coefficients. Each identity r has a basal coefficient b_r and each incidence $r \in C_z$ a catalytic coefficient k_{rz} , used in both directions, with

$$\varepsilon \leq b_r \leq 4\varepsilon, \quad 4 \leq k_{rz} \leq 16, \quad \varepsilon = \frac{1}{500\,000\,000}. \quad (2.3)$$

In the averaged model the coefficients are $b_r = \varepsilon S_r B_r$ and $k_{rz} = 4S_r H_{rz}$ with S_r, B_r, H_{rz} independent and uniform on $\{1, \frac{3}{2}, 2\}$; the conditional theorems below hold pointwise on the whole box (2.3). An *environment* ω is the rows together with the coefficients.

Count process. For an integer count scale V , a channel with coefficient κ and input multiplicities s_x , including catalyst roles, has propensity

$$\kappa V^{1-m} \prod_x (N_x)_{s_x}, \quad m = \sum_x s_x, \quad (N)_j = N(N-1) \cdots (N-j+1). \quad (2.4)$$

The catalysed channel of (r, z) has inputs $u + v + z$ and outputs $uv + z$; repeated substrate and catalyst roles are combined before the falling factorial is taken. This is an ordered-encounter coefficient convention without factorial divisors; other conventions absorb such factors into κ [5]. Each food species arrives at rate V and every molecule washes out at rate one per copy, so time is measured in residence times. Initially every food count is V and every nonfood count is zero. On every bounded interval the process is nonexplosive and visits finitely many states, because internal reactions preserve monomer mass and feed arrivals are Poisson [5, 11, 23]. $\mathbb{P}_\omega$ denotes the trajectory law in a fixed environment. Normalized coordinates are $x_z = N_z/V$, the total mass $L = V^{-1} \sum_z |z| N_z$, the nonfood mass $m = V^{-1} \sum_{|z| \geq 2} |z| N_z$, and we write

$$u = x_{00}, \quad w = x_{11}, \quad K = N_{0011} \text{ (a count, not a concentration)}. \quad (2.5)$$

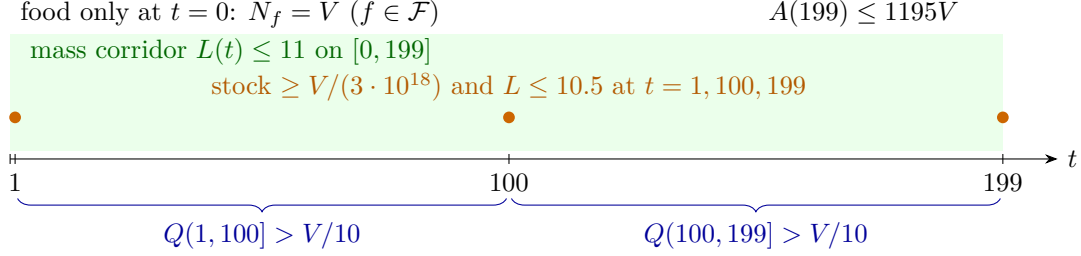


Figure 1: The single-trajectory requirement $F_{n,V}$. Stock is observed at three times; the corridor and the supply allowance cover the whole horizon, including startup from the food-only state.

2.2 The operating requirement

Let $Q(s, t]$ be the monomer equivalents removed by nonfood washouts during $(s, t]$, $A(t)$ the number of food arrivals by time t , and $B(t) \leq 2A(t)$ their monomer equivalents.

Definition 2.1 (Two-window requirement [4]). The event $F_{n,V}$ requires, on one trajectory:

- (M1) at each of the times 1, 100 and 199, $L \leq \frac{21}{2}$ and some species of length four has at least $V/(3 \times 10^{18})$ copies;
- (M2) $Q(1, 100] > V/10$ and $Q(100, 199] > V/10$;
- (M3) $L(t) \leq 11$ for all $0 \leq t \leq 199$, and $A(199) \leq 1195V$.

There is no reset, resampling of chemistry, or fresh initial law at time 100. The stock condition is existential over length-four species; the proofs follow the single species 0011. Figure 1 displays the requirement.

Lemma 2.2 (Material identity [4]). *Let $J(t)$ be the signed nonfood monomer mass produced by internal reactions up to time t . On every trajectory from the food-only state,*

$$m(t)V + Q(0, t] = J(t), \quad (2.6)$$

and the total monomer supplied by time t is $10V + B(t) \leq 10V + 2A(t)$. [Lean]

Proof. A food arrival does not change nonfood mass; a nonfood washout lowers it by exactly what Q records; an internal reaction changes it by exactly what J records. Summing over the finitely many jumps on $[0, t]$ and using $m(0) = 0$ gives (2.6). $\square$

2.3 The witness class

Let $r_* = (00, 11, 0011)$ and let A_n be the source event that all six food rows are empty and $r_* \in C_{0011}$. No other entry of C_{0011} and no other nonfood row is restricted. Row independence and exchangeability within a row give

$$\mathbb{P}(A_n) = e_\alpha^6 p_{\alpha,n}. \quad (2.7)$$

On A_n the product 0011 catalyses its own formation from two distinct food dimers; this is a self-catalytic ligation witness, not a statement about collectively autocatalytic cycles.

2.4 Static channel fields

A *split channel* is a pair (w, i) with $1 \leq i < |w|$, representing $u + v \rightleftharpoons w$ with $|u| = i$. The *quotient* convention identifies (u, v) with (v, u) exactly when $uv = vu$, and never identifies channels with different products [1]. For $\nu \in \{\text{sp}, \text{qt}\}$ give every channel of all finite lengths an independent Bernoulli(a) open mark. The *ordinary reversible closure* of an open set is the smallest set of words containing $\mathcal{F}$ such that, for every open channel $u + v \rightleftharpoons uv$, availability of u and v implies availability of uv and availability of uv implies availability of u and v . Availability is set-valued; there are no counts, rates or energies. The *survival profile* is

$$S_\nu(a) = \mathbb{P}_a\{\text{the closure contains words of arbitrarily large length}\}. \quad (2.8)$$

In the independent-catalysis binary polymer model with $p_n R_n^\nu/n \rightarrow \lambda$, the probability that a RAF exists converges to $\Theta_\nu(\lambda) = S_\nu(1 - e^{-\lambda})$, with $0 < S_\nu(a) < 1$ for $0 < a < 1$ [1, 3]. The static field is thus the object that determines RAF *availability* in the critical window; it is not the capped-Zipf source of Section 2.1.

3 Main results

Put $V_*(n) = \max\{10^{24}, 10^{13}n\}$ and

$$q(d) = \frac{\frac{19}{10}d - 82}{624} - \frac{1}{20}. \quad (3.1)$$

Theorem 3.1 (Finite startup). *Let $n \geq 4$, $\alpha > 1$ and let $V \geq V_*(n)$ be an integer.*

- (a) *For every environment with rows in A_n and coefficients in (2.3), $\mathbb{P}_\omega(F_{n,V}) \geq 9989/10000$.*
- (b) *The fully averaged success probability satisfies $S_{\alpha,n,V} := \mathbb{P}(F_{n,V}) \geq \frac{9989}{10000} e_\alpha^6 p_{\alpha,n}$.*
- (c) *For every source configuration and every kinetic mark configuration, almost surely on $F_{n,V}$, $J(199) > V/5$.*

[Lean: StartupMarked.startup_quantitative_resolution]

Theorem 3.2 (All windows on one event). *Under the hypotheses of Theorem 3.1(a), let $E_{n,V}$ be the event that $F_{n,V}$ holds and that $Q(s, t] > q(t - s)V$ for every $1 \leq s \leq t \leq 199$. Then the outer probability of the complement of $E_{n,V}$ is less than $11/10000$.¹ Almost surely on $E_{n,V}$, $J(199) > 6V/25$.*

[Lean: StartupMarked.startup_enhanced_failure, RandomViability.physical_enhanced_accounts]

Corollary 3.3 (Output, recovery and deadlines; conventional, from Theorem 3.2 and Lemma 2.2). *Almost surely on $E_{n,V}$:*

- (a) $Q(1, 100]$ and $Q(100, 199]$ each exceed $\frac{749}{6240}V > 0.12V$;
- (b) $Q(1, 199] > \frac{263}{624}V > 0.42V$ and $J(199) > \frac{263}{624}V$;
- (c) $Q(0, 199]/(10V + B(199)) > 263/1497600 > 1/5700$;
- (d) *every window $(s, t] \subseteq (1, 199]$ with $t - s > 1756/19 \approx 92.42$ exports more than $V/10$; in particular the schedule 1, 94, 187 gives two such windows, and no three disjoint windows of that length fit in $[1, 199]$.*

¹The event quantifies over a continuum of windows. The bound is proved for the outer measure of its complement, so no separate measurability argument is needed.

a	$(\lambda = -\log(1-a))$	$S_{\text{sp}}(a)$	$S_{\text{qt}}(a)$	lower-endpoint certificate
9/50	(0.19845)	[0.4748617, 0.9982541]	[0.4568760, 0.9974034]	three-type
1/5	(0.22314)	[0.8753628, 0.9992078]	[0.8606719, 0.9987621]	three-type
1/4	(0.28768)	[0.9957216, 0.9998996]	[0.9941957, 0.9998215]	three-type
3/10	(0.35667)	[0.9998233, 0.9999890]	[0.9997091, 0.9999775]	three-type
		[0.87, $1 - 0.7^{32}$]	[0.87, $1 - 0.7^{30}$]	dimer $r = \frac{22}{25}$ [Lean]
1/2	(0.69315)	[$1 - 200^{-4}$, $1 - 2^{-32}$]	[$1 - \ell_{1/2}$, $1 - 2^{-30}$]	three-type
		[0.9999, $1 - 2^{-32}$]	[0.9999, $1 - 2^{-30}$]	dimer $r = \frac{11}{20}$ [Lean]

Table 2: Certified enclosures of the static survival profiles. Decimal endpoints are rounded outward from exact rational certificates; upper endpoints are the gateway bounds of Theorem 3.4(a). Here $\ell_{1/2} = (53/400)^2(11/160)^6$. These are rigorous enclosures, not statistical intervals. The parenthesized value is the critical-window intensity at which $\Theta_\nu(\lambda) = S_\nu(a)$.

Theorem 3.4 (Static certificates). *Let $\nu \in \{\text{sp}, \text{qt}\}$ and $a \in [0, 1]$, and put $b = 1 - (1 - a)^2$ and $g_{\text{sp}} = 32$, $g_{\text{qt}} = 30$.*

- (a) (Gateways.) $S_\nu(a) \leq 1 - (1 - a)^{g_\nu}$. [Lean]
- (b) (Dimer forest.) *If $0 \leq r \leq 1$ and $1 - a + ar^4 \leq r$, equivalently $a(1 + r + r^2 + r^3) \geq 1$ when $r < 1$, then $S_\nu(a) \geq 1 - r^{16}$.* [Lean]
- (c) (Three-type prefix process.) *If $(x, y, z) \in [0, 1]^3$ satisfies, componentwise,*

$$\left([(1-b)z + bx]^2, [(1-a)z + ax]^2, [1 - a + ay]^2 \right) \leq (x, y, z), \quad (3.2)$$

then

$$S_{\text{sp}}(a) \geq 1 - x^4, \quad S_{\text{qt}}(a) \geq 1 - [(1-a)z + ax]^2 [(1-b)z + bx]^6. \quad (3.3)$$

[conventional]

Table 2 lists certified enclosures. At $a = 1/4$ both profiles exceed 99%, so by monotonicity of Θ_ν [1, 3] the limiting RAF probability in the independent-catalysis binary polymer model exceeds 0.9957 (split) and 0.9941 (quotient) for every intensity $\lambda \geq \log(4/3)$. The dimer construction alone is critical at $a = 1/4$ and certifies nothing there.

4 Proof of the finite startup theorems

Throughout this section ω has rows in A_n and coefficients in (2.3), $n \geq 4$, and $V \geq V_*(n)$. The argument has four parts: a stock floor (Section 4.3), a retained reward whose drift is positive when nonfood mass is small (Section 4.4), a concentration bound for that reward (Section 4.5), and a deterministic occupation argument (Section 4.6). Section 4.7 joins them on one trajectory.

4.1 Why the earlier scale was large

The companion proof [4] compared the count process with its compensator in the concentration coordinates (u, w, x_{0011}) and required $|x_{0011} - \tilde{x}_{0011}| \leq \eta_P = (3 \times 10^{18})^{-1}/100$, one percent of the stock floor. A Freedman-type bound [13] for this coordinate has exponent proportional to $\eta_P^2 V$, so reliability forced $V \gtrsim \eta_P^{-2} \approx 10^{41}$ times constants, about 5×10^{49} . The noise budget was symmetric: favourable product births, which help the stock requirement, were charged exactly as losses. The new proof never asks the product concentration to be close to anything. It asks for a lower count

floor, which only adverse jumps can threaten, and it measures growth through a logarithm whose favourable increments may be discarded.

4.2 Inherited envelopes and controls

Lemma 4.1 (Kinetic envelopes [4]). *At every state with $L \leq 11$:*

- (E1) *every channel's total coefficient, basal plus catalytic contributions weighted by catalyst concentrations, is at most $C = 4\varepsilon + \frac{16}{3}m < 59$;*
- (E2) *the drifts of u and w satisfy $b_u \geq 1 - u - 23Cu$ and $b_w \geq 1 - w - 23Cw$;*
- (E3) *every jump changes K by at most two copies downward, and the copy-weighted loss rate satisfies $\sum_{\Delta K < 0} |\Delta K| \lambda \leq (1 + 25C)K \leq 1476K$;*
- (E4) *every jump changes u and w by at most $2/V$ each, and the total rate of all channels is at most $24000V \leq 96000V$.*

These bounds use only the empty food rows, the coefficient box and the corridor: every active catalyst is nonfood, so its concentration is at most $m/3$; falling factorials are bounded by powers; a species is consumed by ligation in two input positions against partners of total concentration at most 11, and by cleavage at one or three split positions. The total rate bound is the formal statement `unbounded_total_rate_mass_eleven`.

Lemma 4.2 (Controls). *Let $\mathcal{C}$ be the event that, up to time 200 and before the first exit from $L \leq 11$, the compensated total mass stays within $\frac{1}{4}$, the compensated u and w stay within 10^{-5} , the compensated export counter Q/V stays within $\frac{1}{40}$, and the compensated arrival counter A/V stays within 1. Then $\mathbb{P}_\omega(\mathcal{C}^c) \leq 22e^{-79} < 10^{-3}$. On $\mathcal{C}$, for all $t \leq 199$: $L(t) \leq 10 + 2 \cdot \frac{1}{4} = \frac{21}{2}$; $u, w \geq \ell := \frac{1}{1358} - 2 \cdot 10^{-5}$; and $A(199) \leq 1195V$. [Lean: `retained_controls_uniform`, `source_guard_prefix`]*

Proof. The inherited exponential martingale bounds [4] give, for tolerances d_N, d_M, d_F, d_E, d_B satisfying the jump margins $d_N + n/V \leq \frac{1}{8}$, $d_M + 2/V \leq \frac{1}{80}$, $d_F + 2/V \leq 10^{-5}$, $d_E + n/V \leq \frac{1}{40}$ and $d_B + n/V \leq 1$,

$$\mathbb{P}_\omega(\mathcal{C}^c) \leq 2e^{-E_N(d_N)} + 12e^{-E_C(d_M)} + 4e^{-E_C(d_F)} + 2e^{-E_N(d_E)} + 2e^{-E_N(d_B)},$$

with $E_N(s) = s^2V/[4n(96011 \cdot 200 + s)]$ and $E_C(s) = s^2V/[4(96000 \cdot 200 + 2s)]$. The mass tolerance is composed from a nonfood tolerance $\frac{1}{8}$ and six food tolerances $\frac{1}{80}$ weighted by food lengths summing to 10. Take each d equal to 0.99 times its target. The margins hold because $n/V \leq 10^{-13}$ and $2/V \leq 2 \cdot 10^{-24}$. At $V = 10^{24}$ and $V/n = 10^{13}$ the five exponents are 1994, 1.99×10^{12} , 1.28×10^6 , 79.75 and 1.28×10^5 , all above 79 and increasing in V and V/n . Hence the bound is at most $22e^{-79} \approx 1.1 \times 10^{-33}$.

For the deterministic consequences write the mass equation $L(t) = 10 + \int_0^t (10 - L) dv + Z_L(t)$; the compensator is not constant, because dilution contributes $-L$. If $|Z_L| \leq \eta$ then $e = L - 10$ solves $e(t) = Z_L(t) - \int_0^t e^{-(t-v)} Z_L(v) dv$, so $|L - 10| \leq \eta(2 - e^{-t}) \leq 2\eta$. For u , (E1)–(E2) give $b_u \geq 1 - 1358u$, and the same comparison from $u(0) = 1$ with noise at most η_F gives $u(t) \geq \frac{1}{1358} + (1 - \frac{1}{1358})e^{-1358t} - \eta_F(2 - e^{-1358t}) \geq \frac{1}{1358} - 2\eta_F$. Arrivals have compensator $6t$. A first-exit argument removes the stopping at the corridor: the estimates preclude a first violation before time 200. $\square$

The exponent 79.75 for the export counter is the smallest; it is where the linear term $10^{13}n$ in $V_*(n)$ comes from (Section 4.8).

4.3 A guarded count floor

Let $\mathbf{g}$ be the guard $\{L \leq 11, u \geq \ell, w \geq \ell\}$, and write $h = \lceil V/(3 \times 10^{18}) \rceil$ for the stock threshold in counts.

Lemma 4.3 (Power tilt). *Let $q = 9/10$, $c = V/10^{17}$ and $1 \leq m_0 \leq V/(2 \times 10^{18})$ an integer with $m_0 \geq 10$. At every state satisfying $\mathbf{g}$, the generator $\mathcal{L}$ of the count process satisfies*

$$\mathcal{L} q^K \leq -c q^K + \frac{1558}{9} m_0 q^{m_0}. \quad (4.1)$$

[Lean: physical_power_affine_generator]

Proof. A jump changing K by d changes q^K by $q^K(q^d - 1)$. Upward jumps contribute nonpositive terms; retain only the basal forward firing of r_* , whose inputs 00 and 11 are distinct, with rate $b_{r_*} N_{00} N_{11}/V \geq \varepsilon \ell^2 V$ and weight $q - 1 = -\frac{1}{10}$. A loss of $d \in \{1, 2\}$ copies has weight $q^{-d} - 1$, and per copy lost $(q^{-1} - 1) = \frac{1}{9} < \frac{19}{162} = \frac{1}{2}(q^{-2} - 1)$. By (E3), $\mathcal{L} q^K \leq (-\frac{1}{10} \varepsilon \ell^2 V + \frac{19}{162} \cdot 1476 K) q^K = (-\iota + \frac{1558}{9} K) q^K$ with $\iota = \varepsilon \ell^2 V/10$. If $K \leq m_0$, then $\frac{1558}{9} K q^K \leq \frac{1558}{9} m_0 q^K$; if $K > m_0$, then $K q^K \leq m_0 q^{m_0}$ because $k q^k$ decreases for $k \geq 10$. In both cases (4.1) follows from $c + \frac{1558}{9} m_0 \leq \iota$, that is, $10^{-17} + \frac{1558}{9} \cdot \frac{1}{2 \times 10^{18}} \leq \varepsilon \ell^2/10$, which holds with margin 6.08×10^{-18} . $\square$

Lemma 4.4 (Guarded tail). *Let $m_0 = \lfloor V/(2 \times 10^{18}) \rfloor$. From any initial population, for every $t \geq 1$ and every integer $j \leq m_0$,*

$$\mathbb{P}\{\mathbf{g} \text{ holds at every jump state up to } t, K(t) \leq j\} \leq 10 q^{m_0-j}. \quad (4.2)$$

[Lean: source_count_tail]

Proof. Kill the process at the first jump into a state violating $\mathbf{g}$. Before killing only finitely many states are reachable on $[0, t]$, so the killed semigroup obeys Kolmogorov's equations. Put $f(t) = \mathbb{E}[\mathbf{1}_{\{\text{not killed by } t\}} q^{K(t)}]$. Killing removes a nonnegative payoff, so (4.1) gives $f' \leq -cf + \frac{1558}{9} m_0 q^{m_0}$ and hence the affine envelope

$$f(t) \leq e^{-ct} q^{K(0)} + (1 - e^{-ct}) b, \quad b = \frac{1558 m_0 q^{m_0}}{9c} \leq \frac{1558}{9} \cdot \frac{10^{17}}{2 \times 10^{18}} q^{m_0} < 8.66 q^{m_0}.$$

The formal proof reaches the same envelope through uniformization and a Poisson mixture of the iterated finite kernel, retaining the factor $1 - e^{-ct}$. For $t \geq 1$, $e^{-ct} \leq e^{-V/10^{17}} \leq q^{m_0}$, so $f(t) \leq 10 q^{m_0}$. On the event in (4.2), $q^{K(t)} \geq q^j$, and Markov's inequality gives the claim. $\square$

Proposition 4.5 (Uniform guarded floor). *From any initial population, the probability that some state occupied during $[1, 199]$ has $K < h$ while $\mathbf{g}$ held at all earlier jump states is less than 10^{-6} for every real $V \geq 10^{24}$. [Lean: RandomViability.source_count_floor_uniform]*

Proof. A first entry into $\{K < h\}$ during $[1, 199]$ either is already present at time 1 or occurs by a jump after time 1 from a state with $K \in \{h, h+1\}$. The second boundary state is necessary because a jump can remove two copies; Section 8 gives a small chain in which omitting it undercounts crossings by three orders of magnitude. By (E3) the rate of such jumps from a guarded state is at most $1476(h+1)$. The expected number of crossings in $(1, 199]$ is the integral of this rate against the occupation of $\{K \leq h+1\}$ under the guard, and Lemma 4.4 bounds that occupation at each time. Together with the time-one term,

$$\mathbb{P}(\text{floor failure}) \leq 10 q^{m_0-h+1} + 1476(h+1) \cdot 198 \cdot 10 q^{m_0-h-1} \leq 10\{1 + 1476(h+1) \cdot 198\} q^{m_0-h-1}.$$

The integration uses the actual holding intervals, and divergence of jump times covers every time. Let $N = \lfloor V/10^{20} \rfloor \geq 10^4$. Then $m_0 - h - 1 \geq V/(6 \times 10^{18}) - 3 \geq 10N$ and $h + 1 \leq 100(N + 1)$. Since $q^{10} \leq \frac{1}{2}$, the bound is at most $3 \times 10^8(N + 1)2^{-N}$, which decreases in N and is below 2.4×10^{-20} already at $N = 100$. The event does not assume that the guard survives; Lemma 4.2 supplies that. $\square$

At $V \geq 10^{24}$ the floor h is at least 333334 copies, which is the analytic cutoff used next.

4.4 A retained logarithmic reward

Define the potential and, for each jump $N \rightarrow N'$ through channel ch , the reward

$$\Phi(N) = \log K - 4H, \quad H = (1 - u)_+^2 + (1 - w)_+^2, \quad (4.3)$$

$$R(N, \text{ch}) = \mathbf{1}\{\text{ch} = \text{sel or } K' < K\} (\log K' - \log K) - 4(H(N') - H(N)), \quad (4.4)$$

where sel is the self-catalysed forward channel $00+11+0011 \rightarrow 2 \cdot 0011$. The reward keeps the selected self-catalytic increment and every decrease of K , and discards the logarithmic increments of every other favourable product birth. It keeps the full change of the food penalty. Put $R_{K_0} = \mathbf{1}\{K \geq K_0\}R$ with $K_0 = 333334$.

Lemma 4.6 (One-sided pathwise inequality). *If $K > 0$ at every state of a finite path segment, then $\sum R \leq \Phi(N_{\text{end}}) - \Phi(N_{\text{start}})$. [Lean: retained_path_sum_1e]*

Proof. Each discarded logarithmic increment is nonnegative, so each R is at most the corresponding increment of Φ ; the increments telescope. $\square$

Only the observable is thinned. The chemistry is unchanged: every favourable and adverse channel remains in the generator and in every drift estimate below.

Lemma 4.7 (Drift, bracket and jump size). *At every state with $L \leq 11$ and $K \geq K_0$, with $\mathcal{D}R = \sum_{\text{ch}} \lambda_{\text{ch}} R$,*

$$\mathcal{D}R \geq 2 - 468\varepsilon - 624m - \frac{6146}{K} - \frac{3072000}{V} \geq \frac{19}{10} - 624m, \quad (4.5)$$

$$\sum_{\text{ch}} \lambda_{\text{ch}} R^2 \leq \frac{6146K}{(K-2)^2} + \frac{196608000}{V} \leq \Gamma(V) := \frac{6146 \cdot 333334}{333332^2} + \frac{196608000}{V}, \quad (4.6)$$

and $|R| \leq 2/333332 + 32/V < 1$. At $V \geq 10^{24}$, $200\Gamma(V) < 3.6877 < 4$. [Lean: StartupMarked.source_retained_reward_drift, cutoff_reward_bracket, marked_gamma_budget]

Proof. Logarithmic part. The selected channel has three distinct inputs, so its rate is $k u w K \in [4u w K, 121K]$, using $k \in [4, 16]$ and $u w \leq (11/4)^2$ from $2u + 2w \leq L \leq 11$. For $K \geq 4$ and $|d| \leq 2$, $\log(K + d) - \log K \geq d/K - 2d^2/K^2$. The retained first moment is the selected rate minus the copy-weighted loss rate, at least $4u w K - (1 + 25C)K$ by (E3). The retained second moment is at most the selected rate plus twice the copy-weighted loss rate, because $d^2 \leq 2|d|$, hence at most $121K + 2 \cdot 1476K = 3073K$. The logarithmic drift is therefore at least $4u w - 1 - 25C - 6146/K$.

Food part. The function $h(y) = (1 - y)_+^2$ has derivative bounded by 2 in absolute value and 2-Lipschitz derivative, so $h(y) \leq h(y_0) + h'(y_0)(y - y_0) + (y - y_0)^2$. Summing over channels, $-4 \sum \lambda \Delta H \geq 8(1 - u)_+ b_u + 8(1 - w)_+ b_w - 4 \sum \lambda (\Delta u^2 + \Delta w^2)$. By (E2) and $u(1 - u)_+ \leq \frac{1}{4}$, $8(1 - u)_+ b_u \geq 8(1 - u)_+^2 - 46C$. By (E4) the Taylor term is at most $4 \cdot 8V^{-2} \cdot 96000V = 3072000/V$. The food part is therefore at least $8H - 92C - 3072000/V$.

Combination. With $\alpha = (1 - u)_+$ and $\beta = (1 - w)_+$, $uw \geq (1 - \alpha)(1 - \beta) \geq 1 - \alpha - \beta$ and $\alpha \leq 2\alpha^2 + \frac{1}{8}$, so $uw \geq \frac{3}{4} - 2H$. Adding the two parts gives $\mathcal{DR} \geq 3 - 8H - 1 - 25C - 6146/K + 8H - 92C - 3072000/V$, which is (4.5) since $117C = 468\varepsilon + 624m$. At $K \geq K_0$ and $V \geq 10^{24}$ the constant part is at least $1.98156 > \frac{19}{10}$.

Bracket and jumps. On retained jumps $|\log(K + d) - \log K| \leq |d|/(K - 2)$, and every jump changes $4H$ by at most $4 \cdot 2 \cdot (2/V + 2/V) = 32/V$. With $(a + b)^2 \leq 2a^2 + 2b^2$, the logarithmic contribution is at most $2 \cdot 3073K/(K - 2)^2$ and the food contribution at most $2 \cdot 96000V \cdot (32/V)^2$. The map $K \mapsto K/(K - 2)^2$ decreases, which gives Γ . The numerical claims are exact rational comparisons. $\square$

Below the cutoff the analytic reward R_{K_0} vanishes, so (4.6) and $|R_{K_0}| < 1$ hold at every state of the corridor. The cutoff changes only the observable, never the process. On the event of Proposition 4.5, $R_{K_0} = R$ at every jump after time 1.

4.5 Concentration of the retained reward

Let $Z(t)$ be the compensated cutoff reward, $\sum_{\text{jumps} \leq t} R_{K_0} - \int_0^t \mathcal{DR}_{K_0}(N_v) dv$, stopped at the first exit from $L \leq 11$ and at time 200.

Lemma 4.8 (Exponential bound). *For every $\delta > 0$, $\mathbb{P}_\omega\{\sup_{t \leq 200} |Z(t)| \geq \delta + 1\} \leq 2e^{-\delta + 200\Gamma(V)}$. In particular $\mathbb{P}_\omega\{\sup_{t \leq 200} |Z(t)| \geq 15\} \leq 2e^{-10} < 99 \times 10^{-6}$. [Lean: source_cutoff_interval_tail, marked_noise_sharp_scalar]*

Proof. Fix $\theta \in \{1, -1\}$. Since $|\theta R_{K_0}| \leq 1$ and $e^y \leq 1 + y + y^2$ for $|y| \leq 1$, at every corridor state $\sum \lambda(e^{\theta R_{K_0}} - 1) \leq \theta \mathcal{DR}_{K_0} + \Gamma$. Consider one holding interval from a state with total rate Λ . Its length τ is exponential with parameter Λ , and the channel is chosen with probabilities $\lambda_{\text{ch}}/\Lambda$ independently of τ . The one-clock identity

$$\mathbb{E} \exp(\theta R_{K_0} - (\theta \mathcal{DR}_{K_0} + \Gamma)\tau) = \frac{\Lambda + \sum \lambda(e^{\theta R_{K_0}} - 1)}{\Lambda + \theta \mathcal{DR}_{K_0} + \Gamma} \leq 1$$

shows that $\exp(\theta Z - \Gamma \cdot \text{elapsed time})$, sampled at jump times and censored at the corridor exit and at time 200, is a nonnegative supermartingale. The maximal inequality for such supermartingales, applied to the event that the prefix value $\theta Z \geq \delta$ at some jump time, gives probability at most $e^{-\delta + 200\Gamma}$ [8, 13]; the two signs give the factor two. Inside a holding interval, Z moves linearly from one prefix value to the next prefix value minus the jump R_{K_0} , so its absolute value exceeds the larger of the two adjacent prefix values by at most one. Taking $\delta = 14$ and $200\Gamma < 4$ gives the stated bound, and $e > 27/10$ gives $e^{10} > 2\,000\,000/99$. $\square$

4.6 Occupation forces export

Proposition 4.9 (Window occupation and quota). *Let a trajectory have positive holding times and divergent jump times, lie in the event $\mathcal{C}$ of Lemma 4.2, avoid the floor failure of Proposition 4.5, and satisfy $|Z| \leq \delta + 1$ on $[0, 200]$. Then for all $1 \leq s \leq t \leq 199$,*

$$\int_s^t m(v) dv > \frac{\frac{19}{10}(t - s) - 54 - 2\delta}{624}, \quad \frac{Q(s, t]}{V} > \frac{\frac{19}{10}(t - s) - 54 - 2\delta}{624} - \frac{1}{20}. \quad (4.7)$$

At $\delta = 14$ this is $Q(s, t] > q(t - s)V$. [Lean at $\delta = 14$: source_window_occupation_quota, source_window_export_quota]

Proof. By Lemma 4.2 every state occupied on $[0, 199]$ satisfies $\mathbf{g}$, so by Proposition 4.5 every state occupied on $[1, 199]$ has $K \geq h \geq K_0$. Hence $R_{K_0} = R$ on the window and (4.5) applies to every occupied state. The compensation identity over the jumps in $(s, t]$, including the two partial holding intervals, reads

$$\sum_{(s,t]} R = \int_s^t \mathcal{D}R(N_v) dv + Z(t) - Z(s) \geq \frac{19}{10}(t-s) - 624 \int_s^t m dv - 2(\delta+1).$$

By Lemma 4.6 the left side is at most $\Phi(N_t) - \Phi(N_s)$. At time t , $4K \leq LV \leq 11V$ gives $\Phi \leq \log(11V/4)$. At time s , $K \geq V/(3 \times 10^{18})$ and $H \leq 2$ give $\Phi \geq \log(V/(3 \times 10^{18})) - 8$. So $\Phi(N_t) - \Phi(N_s) \leq \log(\frac{11}{4} \cdot 3 \times 10^{18}) + 8 < 51.56 < 52$, which gives the first inequality. Each nonfood molecule of length $|z|$ washes out at rate N_z and removes $|z|$ monomer equivalents, so the compensator of Q/V is exactly $\int m dv$. Two prefix errors of at most $\frac{1}{40}$ give the second inequality. $\square$

4.7 Assembly

Proof of Theorems 3.1 and 3.2. Let G be the intersection of $\mathcal{C}$, the complement of the floor failure, and $\{|Z| \leq 15\}$. Holding times are almost surely positive, since every state has total rate at least $6V$, and jump times diverge. Almost surely on G , Proposition 4.9 with $\delta = 14$ gives $Q(s, t] > q(t-s)V$ for all windows. Here $q(99) = 749/6240 > 1/10$, which gives (M2). Lemma 4.2 gives (M3) and the mass part of (M1), and Proposition 4.5 the stock part of (M1) at times 1, 100 and 199, since each deterministic time lies in a holding interval. By the union bound, with no independence used, the outer probability of the complement of $E_{n,V}$ is less than $10^{-3} + 10^{-6} + 99 \times 10^{-6} = 11/10000$. The measurable event $F_{n,V}$ therefore has probability at least $9989/10000$, which is Theorem 3.1(a). Part (b) follows by integrating (a) over the mark law and over the source restricted to A_n , whose exact mass is (2.7). Part (c) and the synthesis statement of Theorem 3.2 follow from Lemma 2.2: on $F_{n,V}$, $J(199) \geq Q(1, 100] + Q(100, 199] > V/5$, and on $E_{n,V}$ the two quotas exceed $749V/6240$ each, and $749/3120 > 6/25$. $\square$

4.8 What sets the new scale

Each of the two terms of $V_*(n)$ is set by one inequality.

The constant 10^{24} . The bracket (4.6) needs $200 \cdot 6146K/(K-2)^2 < 4$, that is, $K \geq 307304$. The drift needs only $K \geq 61461$. The stock threshold of the unchanged requirement is $V/(3 \times 10^{18})$ copies. For the floor of Proposition 4.5 to reach the cutoff one needs $V \gtrsim 9.2 \times 10^{23}$; with $K_0 = 333334$ this is $V > 9.99999 \times 10^{23}$. Thus 10^{24} is the product of the requirement's own relative stock threshold and the variance constant of the logarithmic reward. Accepting a larger failure budget changes this little: at 99% reliability one may allow $200\Gamma \leq 8.5$, which needs $K \geq 144616$ and lowers the scale only by a factor of about 2.1. A substantially smaller scale needs a different estimate below 3×10^5 copies, not a retuning of this one.

The term $10^{13}n$. The inherited export-counter control has exponent $d_E^2 V/[4n(96011 \cdot 200 + d_E)]$ with $d_E = 0.99/40$. It exceeds 79 exactly when $V/n \geq 9.906 \times 10^{12}$. Mass and arrival counters jump by up to n/V , so their exponents carry a factor $1/n$; the coordinate controls do not.

5 Consequences

5.1 Output, recovery and deadlines

Proof of Corollary 3.3. (a) and (b) are $q(99) = 749/6240$ and $q(198) = 263/624$ in Theorem 3.2, applied to $(1, 100]$, $(100, 199]$ and $(1, 199]$. By (2.6), $J(199) = m(199)V + Q(0, 199] \geq Q(1, 199]$. (c)

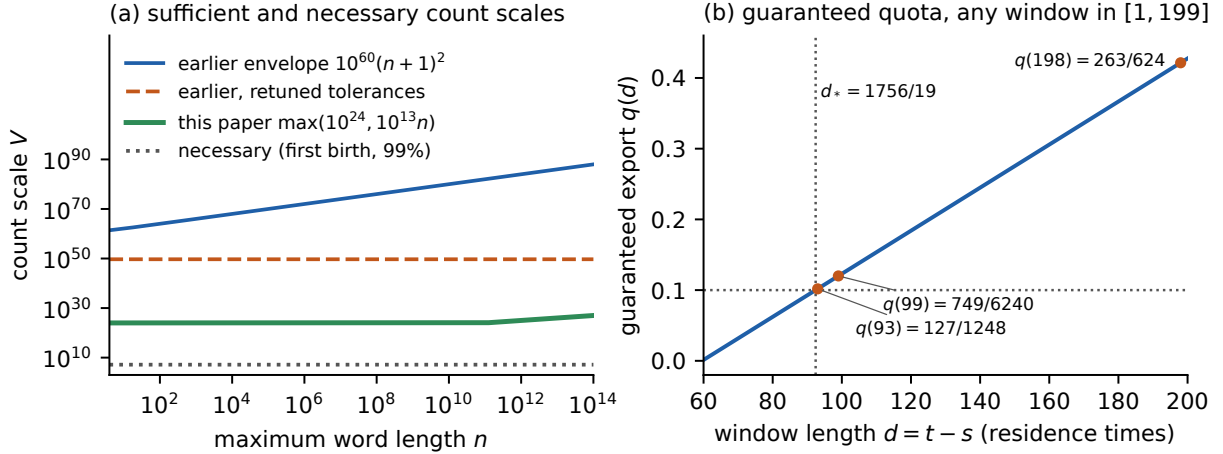


Figure 2: (a) Sufficient count scales for the same two-window requirement: the earlier envelope, the earlier scale after unequal tolerances, and the scale proved here, together with the necessary first-birth scale for 99% reliability in food-silent environments. The curves have different logical roles and are not estimates of a transition. (b) The guaranteed export $q(d)$ for any window of length d inside $[1, 199]$ on the single event of Theorem 3.2.

follows from (b) and $10V + B(199) \leq 10V + 2 \cdot 1195V = 2400V$, since $263/(624 \cdot 2400) = 263/1497600$. For (d), $q(d) > 1/10$ exactly when $d > 1756/19$. Three disjoint windows of that length need more than 277 residence times inside $[1, 199]$. $\square$

The recovery ratio is a guaranteed floor in monomer equivalents, including the initial inventory. It is not a measured or typical conversion efficiency, and nothing here bounds typical recovery from above. Figure 2(b) shows the quota function. Windows shorter than 59.58 residence times receive no guarantee; the drift must first pay for the potential range and the endpoint noise.

5.2 The true failure budget and a reliability trade-off

The formal root rounds the three failure terms up to 10^{-3} , 10^{-6} and 99×10^{-6} . Their proved sizes are much smaller, except for the martingale term.

Proposition 5.1 (Reliability–quota trade-off). *Under the hypotheses of Theorem 3.1(a), for every $\delta > 0$, with outer probability at least $1 - 2e^{-\delta+3.6877} - 1.1 \times 10^{-33} - 10^{-2990}$, every window $(s, t] \subseteq (1, 199]$ exports more than $[(\frac{19}{10}(t-s) - 54 - 2\delta)/624 - \frac{1}{20}]V$. Consequently:*

- (a) *with $\delta = 14$, the event $E_{n,V}$ has outer failure probability below 9.1×10^{-5} ;*
- (b) *with $\delta = 20$, $\mathbb{P}_\omega(F_{n,V}) > 1 - 1.7 \times 10^{-7}$, because windows of length 99 then still export more than $\frac{629}{6240}V > V/10$.*

[conventional]

Proof. Lemma 4.8 holds for every δ . Proposition 4.9 is stated for general δ . The controls fail with probability at most $22e^{-79} < 1.1 \times 10^{-33}$ by the proof of Lemma 4.2. The floor fails with probability at most $3 \times 10^8(N+1)2^{-N}$ with $N \geq 10^4$, below 10^{-2990} . At $\delta = 14$, $2e^{-10} < 9.081 \times 10^{-5}$. At $\delta = 20$, $e^{20-3.6877} > 1.21 \times 10^7$. All comparisons are exact rational computations. $\square$

A larger δ buys reliability at the cost of quota and deadline. With $\delta = 20$ the least window length giving $V/10$ rises from 92.42 to 98.74, so the original windows of length 99 are just admissible. Reliability does not tend to one along this family at a fixed horizon: windows of length 99 with quota $V/10$ force $\delta < 20.25$, so this route cannot push the failure bound for $F_{n,V}$ below about 1.3×10^{-7} .

5.3 Independent screening

At $n = 4$ and $\alpha = 2$, $R_4 = 68$, $e_2 = 6/\pi^2$ and

$$\mathbb{E}D = 67 - e_2 \sum_{k=1}^{67} \frac{68-k}{k^2} \approx 2.5239961939, \quad w := e_2^6 \frac{\mathbb{E}D}{68} \approx 0.001873651491.$$

Rational enclosures of π from Machin's formula with alternating-series remainders certify $\frac{9989}{10000}w > L_0 = 187159/10^8$ and $(1 - L_0)^{1600} < 1/20$. If each trial independently redraws chemistry, coefficients and trajectory, the probability of at least one success in 1600 trials is therefore above 95%. With the weaker reliability 0.99 the same calculation needs 1614 trials and gives only 0.9487 at 1600. Repeated runs in one sampled chemistry are conditionally independent given that chemistry but dependent after averaging, and the calculation does not apply to them. These are lower bounds for a screening design, not predicted success frequencies or recommended experiments.

5.4 Rarity along the critical sequence

Along $\alpha_n = 2 - 2/n$, $e_{\alpha_n} \rightarrow 6/\pi^2$, $\mathbb{E}D/n \rightarrow 9/\pi^2$ and $R_n/(n2^n) \rightarrow 2$, so $2^n p_{\alpha_n, n} \rightarrow 9/(2\pi^2)$. Theorem 3.1(b) then gives, along every integer sequence $V_n \geq V_*(n)$,

$$\liminf_{n \rightarrow \infty} 2^n S_{\alpha_n, n, V_n} \geq \frac{9989}{10000} \left(\frac{6}{\pi^2}\right)^6 \frac{9}{2\pi^2} \approx 2.30 \times 10^{-2}. \quad (5.1)$$

The normalized source limit and the normalized finite lower bound are formal (`startup_witness_normalized_limit`, `startup_normalized_source_lower`); the final conversion by $R_n/(n2^n) \rightarrow 2$ is conventional. The companion converse $S \leq 224p + \bar{r}$ [4] is *not* transferred. Its residual contains a term $Ce^{-cV_n/n}$, which is negligible relative to 2^{-n} only when $cV_n/n - n \log 2 \rightarrow \infty$, and a fixed ratio V_n/n does not guarantee this. Failure of this transfer is not evidence that the converse fails at the new scale; it is simply not proved.

5.5 Weak food catalysis

The empty food rows are a genuine hypothesis. Food catalysis can initiate immediately [4], but it also accelerates food depletion and adverse reactions. Coefficients below 4 lie outside the box (2.3), so weak food catalysis defines a modified kinetic family, not another point of the original one.

Proposition 5.2 (Finite-horizon continuity). *Fix an environment with rows in A_n and coefficients in (2.3). Add, for each food f and each identity $r = (u, v, uv)$, the catalysed channels $u + v + f \rightarrow uv + f$ and $uv + f \rightarrow u + v + f$ with coefficients in $[0, \eta]$ and propensities (2.4). Let $\mathbb{P}_\omega^\eta$ be the resulting law. For every event F determined by the path on $[0, T]$ and contained in $\{L(t) \leq 11, t \leq T\}$,*

$$\mathbb{P}_\omega^\eta(F) \geq e^{-1452\eta VT} \mathbb{P}_\omega(F). \quad (5.2)$$

In particular $\mathbb{P}_\omega^\eta(F_{n,V}) \geq 0.9989 e^{-1452 \cdot 199 \eta V}$. [conventional]

Proof. Use the random time-change representation [5, 11]. Drive every original channel k by an independent unit Poisson process Y_k and every added channel j by an independent unit Poisson process Y'_j . The original process N solves $N(t) = N(0) + \sum_k \zeta_k Y_k(\int_0^t \lambda_k(N_v) dv)$, and the modified process $\tilde{N}$ solves the same equation with the added terms. By induction over jumps, $\tilde{N} = N$ on $[0, T]$ on the event that no Y'_j has a point in $(0, \int_0^T \lambda'_j(N_v) dv]$. On the corridor, $x_f \leq L$ and falling factorials are bounded by powers, so

$$\sum_j \lambda'_j(N) \leq \eta V \sum_{f \in \mathcal{F}} x_f \left(\sum_{(u,v)} x_u x_v + \sum_z (|z| - 1) x_z \right) \leq \eta V \cdot 11 \cdot (121 + 11) = 1452 \eta V.$$

The Y'_j are independent of N . Conditional on N , the no-point event therefore has probability $\exp(-\sum_j \int_0^T \lambda'_j(N_v) dv)$, which is at least $e^{-1452\eta VT}$ on F . On F intersected with that event, $\tilde{N} = N \in F$. $\square$

Proposition 5.2 establishes continuity, but at $V = 10^{24}$ and $T = 199$ a loss below 10^{-3} requires $\eta \leq 3.46 \times 10^{-33}$. It asks for no added reaction at all, whereas a macroscopic reactor tolerates many small extra reactions. A useful robustness theorem would re-derive the drift (4.5) with added food catalysis. On the corridor the channel coefficient envelope grows by at most 11η , which suggests the replacement $2 - 468\varepsilon - 624m - 1287\eta - 6146/K - 3072000/V$. But the copy-loss envelope, the food drift, the bracket and the guard survival all change, and none of these has been re-derived. We state this as an open problem, not a result.

6 Proof of the static certificates

6.1 Gateways

Proof of Theorem 3.4(a). Ligations of two food words whose product is nonfood have products of length three or four. There are 36 ordered food pairs, 4 of which have a food product, leaving 32 split *growth gateways*. Under the quotient, exactly the pairs $(0, 00)$, $(00, 0)$ and $(1, 11)$, $(11, 1)$ merge, leaving 30. If every growth gateway is closed, then ligation from food yields food, cleavage of food yields food, and the closure equals $\mathcal{F}$. The gateway coordinates are independent, which gives $S_\nu(a) \leq 1 - (1 - a)^{g_\nu}$. $\square$

6.2 Dimer forest

Proof of Theorem 3.4(b). Consider the restricted construction that only appends a food dimer to an available word of even length. Index a branch by a sequence of dimers $(d_1, d_2, \dots, d_k)$, $k \geq 2$; its edge is the channel producing $d_1 \cdots d_k$ from $d_1 \cdots d_{k-1}$ and d_k . Parsing into two-letter blocks is unique, so the product word determines the index. Distinct indices therefore have distinct products and use distinct channel coordinates, in the split model and in the quotient model alike, since the quotient never identifies channels with different products. Pulling back the independent field gives a full product law on the forest. It consists of 16 independent four-ary marked trees, one for each first edge $d_1 d_2$, and each edge is open independently with probability a . A tree fails by depth $k + 1$ if its root edge is closed, or if it is open and all four subtrees fail by depth k . Hence the failure probabilities satisfy $r_0 = 0$ and $r_{k+1} = 1 - a + ar_k^4$. Monotonicity of $y \mapsto 1 - a + ay^4$ and $1 - a + ar^4 \leq r$ give $r_k \leq r$ by induction. Forest failure by depth k has probability $r_k^{16} \leq r^{16}$, and these events increase with k , so their union has probability at most r^{16} . On its complement, available words of every even length exist, because every retained step is an allowed ligation of available words through an open channel. This is unbounded closure. The full chemistry is never assumed to be a branching

process; a branching process is embedded in it. Finally $1 - r^4 = (1 - r)(1 + r + r^2 + r^3)$ gives the equivalent form. $\square$

6.3 The three-type prefix process

Proof of Theorem 3.4(c). Construction. Declare every word of length at most two *reached*. A word c with $|c| \geq 3$ is reached if either its prefix c' of length $|c| - 1$ is reached and the channel $(c', \text{last letter})$ is open, or its prefix c'' of length $|c| - 2$ is reached and the channel $(c'', \text{last two letters})$ is open. Both suffixes are food, so every reached word lies in the closure. Give a word c with $|c| \geq 2$ the state $(\text{reached}(c'), \text{reached}(c)) \in \{0, 1\}^2$, and call c *live* if its state is not 00. Write 11, 01 and 10 for the live states. A live word at every length implies reached words of unbounded length.

Coordinates. The two eligible channels of c both have product c , so channels of different words are distinct. For $|c| \geq 4$ the two channels of c are also distinct in the quotient model, since their factor lengths are $\{|c| - 1, 1\}$ and $\{|c| - 2, 2\}$. For $|c| = 3$, $c = \alpha\beta\gamma$, the quotient identifies $(\alpha\beta, \gamma)$ with $(\alpha, \beta\gamma)$ exactly when $\alpha = \beta = \gamma$, that is, for 000 and 111. All channels of words at one length are disjoint from those at every other length.

Branching structure. Given the states of all words of length at most ℓ , the states of the words of length $\ell + 1$ are determined by those states and by fresh coordinates, distinct for distinct children. Each child $c = c'\gamma$ has state $(\text{reached}(c'), \text{reached}(c))$, and $\text{reached}(c)$ depends only on the state of c' and the coordinates of c . Hence, for $|c'| \geq 3$ in both models and for $|c'| \geq 2$ in the split model, the live words form a multitype Galton–Watson process [6, 15] in which each word has two children with independent states given its own:

- from 11, a child is reached with probability $b = 1 - (1 - a)^2$ (state 11), else it has state 10;
- from 01, a child is reached with probability a (state 11), else state 10;
- from 10, a child is reached with probability a through the dimer channel (state 01), else it is dead (00), and dead words have only dead descendants.

Extinction. Let x_k, y_k, z_k be the probabilities that no live word exists k generations below a word of state 11, 01 or 10. Then $(x_0, y_0, z_0) = 0$ and $(x_{k+1}, y_{k+1}, z_{k+1})$ is the left side of (3.2) evaluated at (x_k, y_k, z_k) , since the two children are independent. This map is coordinatewise nondecreasing, so a supersolution (x, y, z) bounds every iterate. Extinction by depth k increases to extinction, whose probability is therefore at most x, y or z .

Initial layers. In the split model the four dimers have state 11 and independent subtrees, so $S_{\text{sp}}(a) \geq 1 - x^4$. In the quotient model start one layer later. The eight words of length three have independent coordinates. Each of the six words other than 000 and 111 is reached with probability b , and 000 and 111 with probability a , because their two descriptions are one coordinate. A reached word has state 11 and an unreached one state 10, since its dimer parent is reached. The length-three subtrees are independent and at depth at least four the recursion is exact, which gives $S_{\text{qt}}(a) \geq 1 - [(1 - a)z + ax]^2[(1 - b)z + bx]^6$. $\square$

The literal restricted construction was also enumerated exactly at $a = 3/10$ to depth two below a dimer: all 2^{12} split assignments and all 2^{11} quotient assignments, including the merged coordinate of 000. The enumerated extinction probabilities agree with the recursion. This is a check of the bookkeeping, not part of the proof.

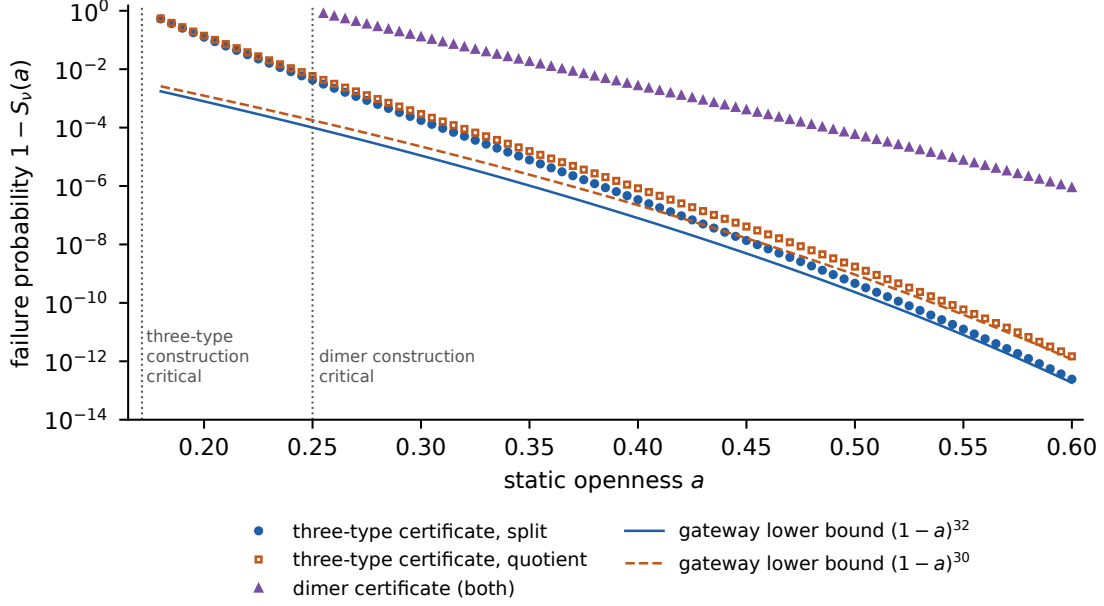


Figure 3: Certified bounds on the failure probability $1 - S_\nu(a)$. Markers are upper bounds from exact three-type and dimer certificates at each plotted openness; lines are the gateway lower bounds. The vertical lines mark where each restricted construction becomes critical; neither is a threshold of the static model, whose survival is positive for every $a > 0$.

6.4 Certificates, inverse design and construction thresholds

A supersolution is found numerically, perturbed along the Perron vector of the Jacobian at the numerical fixed point, rounded up to denominators 10^{12} , and then verified exactly in rational arithmetic. No floating-point fixed point is used as a certificate. Table 2 lists five openness values, Figure 3 shows eighty-five certified points, and Table 3 gives inverse design values. At $a = 1/2$ the short vector $(1/200, 9/500, 13/50)$ has minimum slack $7/25600$ and gives widths below 4×10^{-10} (split) and 10^{-9} (quotient). At $a = 3/10$ the dimer candidates $r = 0.87$ and 0.879 fail the supersolution test while $r = 22/25$ passes; failure of a candidate says nothing about the profile.

The dimer construction has mean offspring $4a$ and is critical at $a = 1/4$. The three-type construction has mean matrix with rows $(2b, 0, 2(1-b))$, $(2a, 0, 2(1-a))$ and $(0, 2a, 0)$, whose spectral radius crosses one at $a \approx 0.1713509469$. Below these values the constructions certify nothing. This is a limitation of the embedded subnetworks, not a threshold of S_ν , which is positive for every $a > 0$ [1, 3]. The companion effective-approximation paper gives an explicit rational evaluator for S_ν at every openness, with an exponentially large cap at moderate a [2]. The certificates here complement it: they are inexpensive and precise from $a \approx 0.2$ upward and silent below.

7 Scale and physical interpretation

Dimensions. For a physical volume Ω in litres, a concentration scale c_* in mol/L and a residence time τ ,

$$V = N_A \Omega c_*, \quad t_{\text{phys}} = \tau t, \quad k_{\text{phys}} = \frac{\kappa}{\tau c_*^{m-1}}, \quad (7.1)$$

Target survival	Necessary: a exceeds		Sufficient: $a \geq$		
	split	quotient	dimer (both)	three-type, split	three-type, quotient
99%	0.1340	0.1423	0.3658	0.2371	0.2412
99.9%	0.1941	0.2056	0.4265	0.2726	0.2792
99.99%	0.2501	0.2643	0.4863	0.3091	0.3183
99.9999%	0.3506	0.3690	0.5972	0.3830	0.3970

Table 3: Inverse design for the static profiles. A necessary entry a_0 satisfies $(1 - a_0)^{g_\nu} > 1 - \text{target}$ exactly, so survival at openness a_0 is below the target. A sufficient entry carries an exact certificate at that openness. These values bracket a target openness without locating any transition.

c_*	1 M	1 mM	1 μ M
Ω at $V = 10^{24}$	1.66054 L	1660.54 L	1.66054×10^6 L

Table 4: Dimensional reading of the count scale. This is a conversion, not a feasible design; the dilute mass-action assumptions are not validated at these concentrations.

where m counts catalyst input roles and $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$ exactly [9]. Table 4 gives the volume at $V = 10^{24}$. Each food species has initial concentration c_* , so the total initial molecular concentration is $6c_*$. A normalized window export of 0.12 is about 0.199 mol of monomer equivalents, and the whole-horizon guarantee 263/624 about 0.700 mol. Meanwhile the guaranteed selected stock is only 333334 molecules, a concentration of $3.3 \times 10^{-19} c_*$. The requirement controls quantities on very different scales, and the export must not be read as the concentration of the selected tetramer. With $\tau = 1$ h the horizon is 199 h. Changing c_* while keeping the dimensionless box fixed changes the physical rate constants, and equal paired dimensionless coefficients have different physical units when their molecularities differ.

Necessary versus sufficient. In every food-silent environment the mission probability is at most $1 - \exp[-(484/3)\varepsilon V]$ [4], so 99% reliability needs $V \geq 14\,272\,222$, about 23.7 pL at 1 μ M. The sufficient scale 10^{24} covers far more than first birth: stock persistence, food and mass control, and two export windows. The gap of seventeen orders of magnitude is unresolved sharpness. Neither endpoint is an observed transition, and Section 4.8 shows that closing the gap requires an estimate below about 3×10^5 copies of the product.

What transfers. The transferable object is a proof contract for a stochastic operating question: (1) specify the chemistry or its source, the initial counts and the coefficient uncertainty; (2) specify stock, output, deadlines, feed and losses as one trajectory event; (3) separate the probability of acquiring the chemistry from the probability of completing the task given it; (4) choose an observable whose fluctuations are exactly the ways the task can fail, and check whether favourable reactions inflate a symmetric noise estimate; (5) verify the count floor, retained drift, bracket, export and mass inequalities for the actual generator; (6) report conditional and averaged guarantees with their units and their unvalidated assumptions. Monomer conservation is a material balance and not a thermodynamic statement; free-energy supply and dissipation need their own accounting [25]. No assay threshold, clinical risk estimate or reactor design follows from the present abstract model.

8 Diagnostics that shaped the proof

The following small computations were run to test proposed inequalities. None is a premise of any theorem.

Symmetric versus retained variance. At $n = 8$, $V = 10^{24}$, all food counts equal to V , $K = 333334$, other nonfood counts zero, and the full catalogue of basal channels, feed and washout, the literal generator gives a retained bracket $\sum \lambda R^2 \approx 1.50 \times 10^{-5}$. The logarithmic increments of favourable non-selected product births have $\sum \lambda (\Delta \log K)^2 \approx 1.80 \times 10^4$. A symmetric fluctuation estimate for $\log K$ is dominated by reactions that help the requirement; discarding them costs only a one-sided inequality (Lemma 4.6). Across ten fixtures at $n \in \{4, 8\}$, including hostile trimer backgrounds with $m = 10$ and repeated-role double losses, every envelope of Lemma 4.7 held.

Two boundary states. In a chain on $\{0, \dots, 30\}$ with immigration rate 8, loss $k \rightarrow k - 2$ at rate $k/2$ ($k \geq 2$) and $1 \rightarrow 0$ at rate 1, and threshold $K < 4$ from $K = 5$, the probability of crossing by time 0.01 is 0.02374. The expected number of crossings from state 4 alone is 1.36×10^{-5} , so omitting the second boundary state $h + 1$ gives an invalid bound. Including it gives 0.02375.

Affine renewal. The naive envelope $\mathbb{E}f \leq e^{-ct}f + b$, iterated over renewal steps, charges b repeatedly and is false: in a small finite-kernel test with $b = 0.2$ and $f = 0$ the true value was 0.20494. The retained form $e^{-ct}f + (1 - e^{-ct})b$ of Lemma 4.4 is what the formal recursion uses.

Logarithm Lipschitz constant. With the denominator $K - 2$ the bracket gives $200\Gamma \approx 3.688$. A gratuitous factor two in the logarithm's Lipschitz constant would exceed the budget 4 that makes $\delta = 14$ yield $2e^{-10}$.

9 Discussion

Availability, initiation and operation. Three requirements are routinely conflated when a structural autocatalytic certificate is read as a prediction. *Availability* of an autocatalytic organization is a source or static-field event: probability $\Theta_\nu(\lambda)$ in the independent critical window, certified here above 99% at $\lambda \geq \log(4/3)$, and $e_\alpha^\delta p_{\alpha,n} = \Theta(2^{-n})$ for the witness class of the capped Zipf source. *Initiation* in a food-silent environment is a first-passage event with hazard at most $(484/3)\varepsilon V$. *Operation*, a trajectory event with stock, deadlines, quotas and a supply account, holds with probability at least 0.9989 given the witness at $V \geq V_*(n)$. The three probabilities live on different spaces and are reported separately.

Why the observable matters. The improvement from 5×10^{49} to 10^{24} did not come from a stronger concentration inequality. It came from deriving the observable from the failure event. Stock can only be lost through adverse jumps, and growth need only be certified from below. A count-tail estimate and a one-sided reward encode exactly these facts, and the fluctuation budget stops paying for favourable reactions.

Open problems. (1) *Smaller scale.* Below 3×10^5 copies the logarithmic bracket fails. A seed-to-established-stock comparison with relative error, jointly with food retention, is needed to approach the necessary scale. (2) *Weak food catalysis.* Re-derive the loss envelope, drift, bracket and guard survival with added food-catalysed channels of strength η , and find the largest η for

which the conclusion of Theorem 3.1 survives at fixed reliability. (3) *Converse at the new scale.* Prove $S_{\alpha_n, n, V_n} = O(2^{-n})$ and singleton selection along sequences with $V_n = O(n)$, or show that they fail. (4) *Collective witnesses.* Replace the self-catalytic ligation by a mutually catalytic pair or cycle; the floor and reward arguments then need multi-species potentials. (5) *Low openness.* Certify $S_\nu(a)$ to useful precision below $a \approx 0.17$, where both embedded constructions are subcritical. (6) *Formalization.* The three-type static bound and the trade-off of Proposition 5.1 are conventional; formalizing them requires a finite-depth joint-law lemma for the live-prefix states and a general- δ occupation statement.

Data and code availability. The repository location will be supplied with the author information. The source package contains this manuscript, its bibliography, the figure script, and an exact-arithmetic script that replays every printed number, certifies every static point and enumerates the literal prefix construction. It also contains the diagnostic scripts and their outputs, and the verification manifest of the formal development (Appendix A).

A Formal scope and reproducibility

The formal development is a single Lean 4 project pinned to `leanprover/lean4:v4.30.0` with a pinned Mathlib manifest. Verification is performed through a wrapper that treats every warning as an error and records a strict receipt with the source hash. Explicit declaration probes report only the standard axioms `propext`, `Classical.choice` and `Quot.sound`; no admitted proof or additional axiom occurs. The physical process is the literal continuous-time jump law with the propensities (2.4), all channels present, and the source is the capped Zipf configuration weight with the uniform finite mark law. Table 5 maps statements to evidence. Declarations live under `proofs/StartupMarked/`, `proofs/StartupCount/`, `proofs/PracticalCriticalWindow/`, `proofs/RepeatedFunction/` and `proofs/RandomViability/`.

Table 5: Statements of this paper and their evidence.

Statement	Evidence	Declarations or remark
Theorem 3.1	Lean	<code>StartupMarked.startup_quantitative_resolution</code> (root; uses 9989/10000, the exact witness mass and the signed synthesis account)
Theorem 3.2	Lean	<code>StartupMarked.startup_enhanced_failure</code> , <code>StartupMarked.source_window_export_quota</code> , <code>RandomViability.physical_enhanced_accounts</code>
Lemma 2.2	Lean	<code>physical_mission_accounts</code> and the companion identity <code>actual_net_synthesis_identity</code>
Lemma 4.1	inherited	companion derivation [4]; <code>unbounded_total_rate_mass_eleven</code> , <code>physical_crossing_rate_le</code>
Lemma 4.2	Lean	<code>retained_controls_failure</code> , <code>retained_controls_uniform</code> , <code>retained_exponents</code> , <code>physical_mass_localization</code> , <code>physical_food_floor_from_noise</code> , <code>source_guard_prefix</code>
Lemma 4.3	Lean	<code>physical_power_generator</code> , <code>physical_power_affine_generator</code> , <code>affine_power_envelope</code>

Statement	Evidence	Declarations or remark
Lemma 4.4	Lean	poissonized_affine_generator,
Proposition 4.5	Lean	source_count_terminal_bound, source_count_tail source_count_crossing_bound, source_count_floor_coarse, count_rounding_bounds, uniform_crossing_scalar_certificate, source_count_floor_uniform
Lemma 4.6	Lean	retained_reward_le_potential_increment, retained_path_sum_le
Lemma 4.7	Lean	source_retained_reward_drift, source_retained_reward_bracket, cutoff_reward_bracket, cutoff_reward_abs, marked_drift_margin, marked_gamma_budget, marked_jump_budget
Lemma 4.8	Lean	gamma_reward_crossing_tail, gamma_reward_two_sided_tail, gamma_reward_interval_tail (general δ), source_cutoff_interval_tail ($\delta = 14$), marked_noise_sharp_scalar
Proposition 4.9	Lean at $\delta = 14$	source_window_occupation_quota, source_window_export_quota; general δ conventional
Section 4.7	Lean	startup_enhanced_mission, startup_mission_failure, startup_concrete_success, startup_fully_averaged_lower
Section 4.8	arithmetic	exact thresholds 307304, 61461, 9.906×10^{12}
Corollary 3.3	conventional	from Theorem 3.2; the 6/25 synthesis bound is Lean
Proposition 5.1	conventional	general- δ tail is Lean; control and floor sizes from their formal proofs
Section 5.3	arithmetic	Machin enclosure; integer powers
Equation (5.1)	Lean + conv.	startup_witness_normalized_limit, startup_normalized_source_lower; final conversion conventional
Proposition 5.2	conventional	random time-change coupling
Theorem 3.4(a)	Lean	split_gateway_upper, quotient_gateway_upper
Theorem 3.4(b)	Lean	dimerPrefixIndex_injective, sourceForest_law, appendTree_generation, dimer_split_survival_lower, dimer_quotient_survival_lower, inverse_source_certificate, half_open_profile_real_intervals, three_tenths_profile_real_intervals
Theorem 3.4(c)	conventional	proof in Section 6; exact enumeration check to depth two
Tables 2, 3; Figure 3	arithmetic	exact rational supersolutions
Section 8	numerical	diagnostics only

Reproduction. The script `check_paper.py` replays every printed number in exact rational arithmetic in about twenty seconds. It checks the floor constants and rounding, the drift, bracket and jump constants, the control exponents, the occupation and quota algebra, the budget and its sharpening, the screening enclosure, the necessary scale, the weak-food constant and the unit conversions. It also searches for, rounds and exactly verifies every static certificate, computes

the inverse table and the construction thresholds, and enumerates the literal prefix construction. The script `make_figures.py` draws both figures from these certified values. Formal verification uses the repository wrapper with an explicit declaration, for example `StartupMarked.startup_quantitative_resolution` in `proofs/StartupMarked/Resolution.lean`. The source hashes of the eight principal files match the verification manifest distributed with the package; no dependency update is performed.

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